

Research Interests

I am fascinated by logic and its applications in theoretical computer science. I am interested in formal verification, having worked with automated reasoning tools and interactive theorem proving. More broadly, I am interested in learning about algorithms, complexity and combinatorics through the lens of logic.

Education

2026 – 2027 **Master of Philosophy in Engineering.**

University of Sydney

Topic: Best-Effort Synthesis for Quantitative Graph Games

Supervisor: Dr Sasha Rubin

2025 – 2025 **Bachelor of Science with Honours in Pure Mathematics.**

University of New South Wales, Honours Class 1, 91.750 WAM

Thesis: Ehrenfeucht-Fraïssé Games for Temporal (and) Epistemic Logics

Supervisors: Dr Paul Hunter and Dr Thomas Britz

2022 – 2024 **Bachelor of Science with Distinction in Mathematics.**

University of New South Wales, 94.087 WAM

Publications

Under Review Characterising Equivalences Between Logics Through Model Comparison Games.
Ronald Chiang, Paul Hunter. *Manuscript under review.*

Research Experience

University of New South Wales

February 26 **Model Comparison Games.**

July 26 Created novel games for branching time, temporal epistemic and transitive closure logics. Proved original inexpressibility results for temporal epistemic logics. Provided various novel proof techniques using games.

Supervisor: Dr Paul Hunter

Sonic Labs

December 24 **Comparing Automated Inductive Invariant Inference Tools for IVy and TLA⁺.**

February 25 Compared the abilities of endive, for TLA⁺, and SWISS, for IVy, to generate inductive invariants. Developed new implementation techniques for threshold and round-based consensus protocols.

Supervisors: Dr Sasha Rubin and Dr Pavle Subotić

Trustworthy Systems

February 24 **Specifying Properties for Device-Driver Verification.**

September 24 Developed a framework to encode properties about Pancake programs in Linear Temporal Logic for HOL4.

Supervisors: Dr Johannes Pohjola and Dr Miki Tanaka

Awards

- 2026 – 2027 **Australian Government Research Training Program Fee Offset.**
2025 **J Holden Family Foundation Honours Award in Maths and Physics.**
2024 **Automated Proofs for TLA⁺ – Vacation Research Internship Program Scholarship.**
2024 **UNSW Engineering Taste of Research Scholarship.**
2022 – 2024 **UNSW Science Dean's List for Academic Excellence (Top 100 in Science).**
2023 **UNSW Engineering Dean's Award for Academic Excellence (Top 100 in Engineering).**

Teaching

USyd Computer Science

COMP2022: Models of Computation, 26S2.

UNSW Computer Science and Engineering

COMP1531: Software Engineering Fundamentals, 23T1, 23T2, 23T3.

COMP2041: Software Construction, 24T1.

COMP3121: Algorithm Design and Analysis, 23T2, 23T3, 24T1, 24T2.

COMP3153: Algorithmic Verification, 25T3.

COMP4141: Theory of Computation, 26T1.

COMP4161: Advanced Topics in Software Verification, 26T3.

COMP9020: Foundations of Computer Science, 24T2, 24T3, 25T1, 26T1.

Course Administrator for COMP9020 in 25T1, 26T1 and COMP4161 in 26T3

UNSW Mathematics and Statistics

Mathematics Drop-In Centre, 24T1, 24T2, 24T3, 26T1, 26T2.

Mathematics Exam Invigilation, 26T1, 26T2.

MATH1011: Fundamentals of Mathematics, 26T1.

MATH1081: Discrete Mathematics, 25T2, 25T3.

MATH1131: Mathematics 1A, 26T1.

MATH1151: Mathematics for Actuarial Studies and Finance 1A, 26T1.

MATH1231: Mathematics 1B, 25T2, 26T1, 26T2.

Extracurriculars

- 2024 **Vice President of Education, UNSW Mathematics Society.**
2023 **Director of Outreach, UNSW Mathematics Society.**
2022 **Subcommittee Member of Education, UNSW Mathematics Society.**
2022 – 2023 **AV Simulations Developer, UNSW Redback Racing.**

Referees

Sasha Rubin, University of Sydney, sasha.rubin@sydney.edu.au.

Paul Hunter, University of New South Wales, paul.hunter@unsw.edu.au.

Thomas Britz, University of New South Wales, britz@unsw.edu.au.